

Other IoT tools and frameworks

There are many other open source IoT tools and frameworks, as listed below.

- **Eclipse Hono:** This tool works for connectivity and protocol support. It provides various APIs for device communication using arbitrary protocols and also supports protocol customisation.
- **Kaa:** This open source platform is for implementing end-to-end use cases for IoT solutions. It offers various features such as device management, data collection, configuration management, messaging, events, notifications, end points, etc.
- **ThingsBoard:** This open source IoT platform facilitates data collection, processing, visualisation, device management, a rule engine, asset management, customisation and integration.

Some other IoT open source projects are OpenIoT, DeviceHive, Node-RED, IoTivity, Mango and OpenThread.

An IoT solution is an amalgamation of hardware, software and networking capabilities. Emerging technologies and IoT devices will lead to the creation of more IoT applications for both consumers and industrial

use cases. Edge computing is required for any successful and optimised IoT solution. There are various open source frameworks available in the IoT space. Cloud service providers are also providing very rich services for IoT solutions and edge computing. Recommended guidelines for IoT solutions are: loosely coupled, modular, platform-independent, and based on open standards. 

References

- [1] <https://www.lora-alliance.org/technology>
- [2] <https://aws.amazon.com/>
- [3] <https://azure.microsoft.com/en-in/services/>
- [4] <https://www.eclipse.org/kapua/>
- [5] <https://kaaproject.github.io/kaa/docs/v0.10.0/Welcome/>
- [6] <https://thingsboard.io/>

By: Raghendra Singh Dixhit

The author is a technical thought leader and author of the book 'Enterprise Mobility Breakthrough'. He is actively involved with open source frameworks, and has hands-on experience with enterprise integration solutions services, products, etc. He can be contacted at raghu.impros@gmail.com.

Continued from page...98

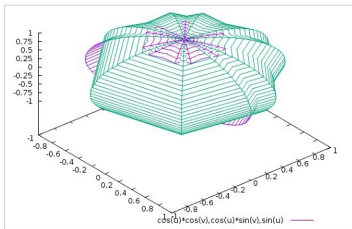


Figure 9: Graph plotted using gnuplot

to bring out a good scientific paper. Do remember that mathematics is full of symbols like π , $\sqrt$, ∞ , etc, which are normally not available on the keyboard. Preparing scientific papers in the required format of various scientific journals is a very tedious task due to reasons like this. So, every mathematician will eventually use a document preparation system and the best one is LaTeX. It is widely used in academia and the industry, to prepare scientific documents in areas like mathematics, statistics, computer science, engineering, etc. LaTeX is cross-platform, free and open source software licensed under the LaTeX

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left[\frac{-\hbar^2}{2\mu} \nabla^2 + V(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t)$$

Figure 10: Schrödinger equation with LaTeX

Project Public License. It can also be used for efficient reference management. Figure 10 shows the time-dependent Schrödinger equation rendered using LaTeX.

Before concluding the article, we need to make a confession —like all Top Ten lists, this too is coloured by personal preferences and prejudices. But we have tried to make the list as diverse and useful as possible so that not just students and practitioners of mathematics, but professionals in computer science, physics, chemistry, etc, can also appreciate and use these tools to their advantage. 

By: Deepu Benson and Jerin Thomas

Deepu Benson is a free software enthusiast whose area of interest is theoretical computer science. He maintains a technical blog at www.computingforbeginners.blogspot.in. And he can be reached at deepumb@hotmail.com.

Jerin Thomas currently works as an assistant professor at Amal Jyothi College of Engineering. His area of interest is digital image processing. He can be reached at jerinthomas@amaljyothi.ac.in.